



Date 10 Sep; 18

MTH 621

Sun Mon Tue Wed Thu Fri Sat

Long Questions:-

(Final 2nd)

- * Prove if f is integrable on $[a, b]$ then $|f|$ is integrable.
- * Is continuity implies differentiability? Explain with example.
- * If f is continuous on $[a, b]$, then it is bounded on $[a, b]$.
- * Find $\int_a^b f(x) dx$?
If $f(x) = \begin{cases} 0 & x \text{ is rational} \\ 1 & x \text{ is irrational} \end{cases}$
- * $h(x) = \sin 1/x$ find derivative, by chain rule.
- * $g(x) = x^2$ show uniform continuity.
- * Define lower integral
- * Define GMVT.
- * check $\lim_{x \rightarrow 2^-} f(x) = f(2)$

$$f(x) = \begin{cases} x^2 & 1 \leq x < 2 \\ n+1 & 1 < x \leq 2 \end{cases}$$

MCQ's:-

AHMAD MSC(MATH)
Ahmad2113@gmail.com

$$\rightarrow \lim_{n \rightarrow \infty} 1/n^p = ?$$

$$\rightarrow \sin x / x^{1/4} = ?$$

$$\rightarrow \text{lower sum}$$

$$\rightarrow \text{upper sum formula.}$$

$$\rightarrow \text{Comparison test}$$

$$\rightarrow \sum_{n=1}^{\infty} 1/n^2 = ?$$

$$\rightarrow \sup_{n \rightarrow \infty} S_n = \infty ?$$

$$\rightarrow e^x \rightarrow \text{in form of}$$

$$\int_a^b f(x) dx = ?$$

Good luck!
:)



5- A theorem was from
Monotonic Function? 5 marks
6- evaluate $\lim_{x \rightarrow 0} \ln \sin x / \ln x$??
7- and Rest two was from the
last 5th chp of about integration
on closed interval. My Yesterday
paper;

Mth621



AHMAD MSC(MATH)

Areeb2113@gmail.com

Mcqs were based on concepts
and 6,7 was from mcqs file of
quizz.

1- Define Removeable
continuity? 2 marks

2- Define chain rule for
composition of functions? 2
marks

3- Define Second mean theorem
of Integration?

4- Find uniform continuity of $2x$?

5- A theorem was from
Monotonic Function? 5 marks

6- evaluate $\lim_{x \rightarrow 0} \ln \sin x / \ln x$??

7- and Rest two was from the



Type a message





Q. prove continuity for the given function. 2 questions

Q. second mean value theorem statement

Q using mean value theorem prove that $x^{-1}/x < \ln x < x-1$ for $x > 1$

Q if $F'(x)=f(x)$ prove integral $f(x)=F(a)-F(b)$ for $[a,b]$

AHMAD MSC(MATH)
Aresh2113@gmail.com

Q if g is continuous at x_0 and f is continuous on $g(x_0)$ then show that $f \circ g$ is continuous at x_0 .

-





AHMAD MSC(MATH)
Areeb2113@gmail.com

Q. No. 1 Prove that if f is continuous on a finite closed Interval $[a, b]$ then f is bounded on $[a, b]$ (5)

Q. No. 2 Suppose that f is monotonic on (a, b) and define $\alpha = \inf f(x)$ If f is nondecreasing ^{as $x \rightarrow b$} then prove that $\lim_{x \rightarrow b} f(x) = \alpha$. (5)

Q. Define a Continuously differentiable function

Q. If Evaluate $\lim_{x \rightarrow \infty} \frac{\ln \sin x}{\ln x}$

Q. Show that the function $g(x) = x^2$ is uniformly continuous on $[-\delta, \delta]$ for any finite δ .

Q. State Generalize mean value theorem



THEOREM: If f is integrable on $[a, b]$ and $[b, c]$, then f is integrable on $[a, c]$.

$$\int_a^c f(x) dx = \int_a^b f(x) dx + \int_b^c f(x) dx. \quad (5.37)$$

Proof: See Lecture.

AHMAD MSC(MATH)
Ahmad2113@gmail.com

5.4 Fundamental Theorem of Calculus

Theorem: If f is integrable on $[a, b]$ and $a \leq c \leq b$, then the function F defined by

$$F(x) = \int_c^x f(t) dt$$

satisfies a Lipschitz condition on $[a, b]$, and is therefore continuous on $[a, b]$.

Proof: If x and x' are in $[a, b]$, then

$$F(x) - F(x') = \int_c^x f(t) dt - \int_c^{x'} f(t) dt = \int_{x'}^x f(t) dt.$$

Since $|f(t)| \leq K$ ($a \leq t \leq b$) for some constant K ,

$$\left| \int_{x'}^x f(t) dt \right| \leq K|x - x'|, \quad a \leq x, x' \leq b.$$





AHMAD MSC(MATH)
Areeb21113@gmail.com

3.5. Continuity

85

Example: Find

$$\lim_{x \rightarrow \infty} \frac{2x^2 - x + 1}{3x^2 + 2x - 1}.$$

Example: Find

$$\lim_{x \rightarrow 1} \frac{x^8 - 1}{x^4 - 1}.$$

$$f(x) = \begin{cases} \frac{1}{x+2}, & x < -2, \\ x^2 - 5, & -2 < x \leq 3, \\ \sqrt{x+13}, & x > 3. \end{cases}$$

3.5 Continuity

Continuity (Informal):

- We say that f is *continuous at* x_0 if f is defined on an open interval (a, b) containing x_0 and

$$\lim_{x \rightarrow x_0} f(x) = f(x_0).$$

- We say that f is *continuous from the left at* x_0 if f is defined on an open interval (a, x_0) and

$$\lim_{x \rightarrow x_0^-} f(x) = f(x_0).$$

- We say that f is *continuous from the right at* x_0 if f is defined on an open interval (x_0, b) and

$$\lim_{x \rightarrow x_0^+} f(x) = f(x_0).$$

Continuity (Formal):

- A function f is continuous at x_0 if and only if f is defined on an open interval (a, b) containing x_0 and for each $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|f(x) - f(x_0)| < \varepsilon \quad \text{whenever} \quad |x - x_0| < \delta.$$

- A function f is continuous from the right at x_0 if and only if f is defined on an interval $[x_0, b)$ and for each $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|f(x) - f(x_0)| < \varepsilon \quad \text{whenever} \quad x_0 \leq x < x_0 + \delta.$$



3.1. Limits

80

First observe that if $L_2 \neq 0$, there is a $\delta_3 > 0$ such that

$$|g(x) - L_2| < \frac{|L_2|}{2}, \quad |g(x)| > \frac{|L_2|}{2} \quad \text{if } 0 < |x - x_0| < \delta_3.$$

Now suppose that $0 < |x - x_0| < \min(\delta_1, \delta_2, \delta_3)$. Then

$$\begin{aligned} \left| \left(\frac{f}{g} \right)(x) - \frac{L_1}{L_2} \right| &= \left| \frac{f(x)}{g(x)} - \frac{L_1}{L_2} \right| = \frac{|L_2 f(x) - L_1 g(x)|}{|g(x) L_2|} \\ &\leq \frac{2}{|L_2|^2} |L_2 f(x) - L_1 g(x)| \\ &= \frac{2}{|L_2|^2} |L_2 [f(x) - L_1] + L_1 [L_2 - g(x)]| \\ &\leq \frac{2}{|L_2|^2} (|L_2| |f(x) - L_1| + |L_1| |L_2 - g(x)|) \\ &\leq \frac{2}{|L_2|^2} (|L_2| + |L_1|) \varepsilon. \end{aligned}$$

AHMAD MSC(MATH)
Areeb2113@gmail.com

Example: Find

$$\lim_{x \rightarrow 2} \frac{9 - x^2}{x + 1} \quad \text{and} \quad \lim_{x \rightarrow 2} (9 - x^2)(x + 1).$$

Solution: If c is a constant, then $\lim_{x \rightarrow x_0} c = c$, and $\lim_{x \rightarrow x_0} x = x_0$. Therefore

$$\begin{aligned} \lim_{x \rightarrow 2} (9 - x^2) &= \lim_{x \rightarrow 2} 9 - \lim_{x \rightarrow 2} x^2 \\ &= \lim_{x \rightarrow 2} 9 - (\lim_{x \rightarrow 2} x)^2 \\ &= 9 - 2^2 = 5, \end{aligned}$$

and

$$\lim_{x \rightarrow 2} (x + 1) = \lim_{x \rightarrow 2} x + \lim_{x \rightarrow 2} 1 = 2 + 1 = 3.$$

Therefore,

$$\lim_{x \rightarrow 2} \frac{9 - x^2}{x + 1} = \frac{\lim_{x \rightarrow 2} (9 - x^2)}{\lim_{x \rightarrow 2} (x + 1)} = \frac{5}{3}$$

and

$$\lim_{x \rightarrow 2} (9 - x^2)(x + 1) = \lim_{x \rightarrow 2} (9 - x^2) \lim_{x \rightarrow 2} (x + 1) = 5 \cdot 3 = 15.$$

Example: The function

$$f(x) = 2x \sin \sqrt{x}$$

satisfies the inequality

$$|f(x)| < \varepsilon, \quad \text{whenever } 0 < x < \delta = \varepsilon/2.$$





Theorem: If f is integrable on $[a, b]$ and $a \leq c \leq b$, then the function F defined by

AHMAD MSC(MATH)
Areeb2113@gmail.com

$$F(x) = \int_c^x f(t) dt$$

satisfies a Lipschitz condition on $[a, b]$, and is therefore continuous on $[a, b]$.

Proof: Consequently,

$$|F(x) - F(x')| \leq K|x - x'|, \quad a \leq x, x' \leq b.$$

Theorem: If f is integrable on $[a, b]$ and $a \leq c \leq b$, then $F(x) = \int_c^x f(t) dt$ is differentiable at any point x_0 in (a, b) where f is continuous, with $F'(x_0) = f(x_0)$.

If f is continuous from the right at a , then $F'_+(a) = f(a)$. If f is continuous from the left at b , then $F'_-(b) = f(b)$.

Proof: Since

$$\frac{1}{x - x_0} \int_{x_0}^x f(x_0) dt = f(x_0).$$

We can write

$$\frac{F(x) - F(x_0)}{x - x_0} - f(x_0) = \frac{1}{x - x_0} \int_{x_0}^x [f(t) - f(x_0)] dt.$$

From this

$$\left| \frac{F(x) - F(x_0)}{x - x_0} - f(x_0) \right| \leq \frac{1}{|x - x_0|} \left| \int_{x_0}^x |f(t) - f(x_0)| dt \right|.$$

Since f is continuous at x_0 , there is for each $\varepsilon > 0$ a $\delta > 0$ such that

$$|f(t) - f(x_0)| < \varepsilon \quad \text{if} \quad |x - x_0| < \delta$$

and t is between x and x_0 .

Therefore, we have





AHMAD MSC(MATH)
Areeb2113@gmail.com

4.5 Rolle's Theorem

Theorem: Suppose that f is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b) , and $f(a) = f(b)$. Then $f'(c) = 0$ for some c in the open interval (a, b) .

Proof: Since f is continuous on $[a, b]$, f attains a maximum and a minimum value on $[a, b]$. If these two extreme values are the same, then f is constant on (a, b) , so $f'(x) = 0$ for all x in (a, b) .

If the extreme values differ, then at least one must be attained at some point c in the open interval (a, b) , and $f'(c) = 0$.

4.6 The Mean Value Theorem

Theorem: Suppose that f is differentiable on $[a, b]$, $f'(a) \neq f'(b)$, and μ is between $f'(a)$ and $f'(b)$. Then $f'(c) = \mu$ for some c in (a, b) .

Proof: Suppose first that

$$f'(a) < \mu < f'(b)$$

and define

$$g(x) = f(x) - \mu x.$$

Then

$$g'(x) = f'(x) - \mu, \quad a \leq x \leq b,$$

and by our supposition $f'(a) < \mu < f'(b)$, we have

$$g'(a) < 0 \quad \text{and} \quad g'(b) > 0.$$

Since g is continuous on $[a, b]$, g attains a minimum at some point c in $[a, b]$.

Due to Lemma and $g'(a) < 0$ and $g'(b) > 0$, imply that there is a $\delta > 0$ such that

$$g(x) < g(a), \quad a < x < a + \delta, \quad \text{and} \quad g(x) < g(b), \quad b - \delta < x < b.$$

Therefore $c \neq a$ and $c \neq b$. Hence, $a < c < b$, and therefore $g'(c) = 0$, by the fact that if f is differentiable at a local extreme point x_0 then $f'(x_0) = 0$. From $g'(c) = 0$, $f'(c) = \mu$. The proof for the case where $f'(b) < \mu < f'(a)$ can be obtained by applying this result to $-f$.

Theorem: If f is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b) , then

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

for some c in (a, b) .



AHMAD MSC(MATH)
Areeb2113@gmail.com

3.3 Limits at $\pm\infty$

If f is defined on an interval (a, ∞) , then $f(x)$ approaches the limit L as x approaches ∞ , and write

$$\lim_{x \rightarrow \infty} f(x) = L.$$

For each $\varepsilon > 0$, there is a number β such that

$$|f(x) - L| < \varepsilon \quad \text{if} \quad x > \beta.$$

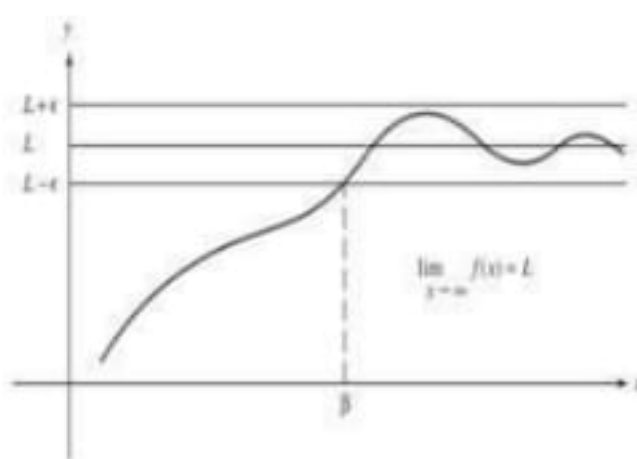


Figure 3.7: Limits at $\pm\infty$

Examples: Let

$$f(x) = 1 + \frac{1}{x^2}, \quad g(x) = \frac{2|x|}{1+x}, \quad \text{and} \quad h(x) = \sin x.$$

Find limits when $x \rightarrow \pm\infty$.

Solution: We have

$$\lim_{x \rightarrow \infty} f(x) = 1,$$

since

$$|f(x) - 1| = \frac{1}{x^2} < \varepsilon \quad \text{if} \quad x > \frac{1}{\sqrt{\varepsilon}}.$$

We have

$$\lim_{x \rightarrow \infty} g(x) = 2,$$

since

$$|g(x) - 2| = \left| \frac{2x}{1+x} - 2 \right| = \frac{2}{1+x} < \frac{2}{x} < \varepsilon \quad \text{if} \quad x > \frac{2}{\varepsilon}.$$

However, $\lim_{x \rightarrow \infty} h(x)$ does not exist, since h assumes all values between -1 and 1 in any semi-infinite interval (τ, ∞) . Discuss the limits at $x \rightarrow -\infty$.

$$f(x) = \frac{1}{x}, \quad g(x) = \frac{1}{x^2},$$





4.11 Taylor's Theorem

Recall the following lemma: If f is differentiable at x_0 , then

$$f(x) = f(x_0) + [f'(x_0) + E(x)](x - x_0),$$

where E is defined on a neighborhood of x_0 and

$$\lim_{x \rightarrow x_0} E(x) = E(x_0) = 0.$$

To generalize this result, we first restate it: the polynomial

$$T_1(x) = f(x_0) + f'(x_0)(x - x_0),$$

which is of degree ≤ 1 and satisfies

$$T_1(x_0) = f(x_0), \quad T_1'(x_0) = f'(x_0).$$

$T_1(x)$ approximates f so well near x_0 that

$$\lim_{x \rightarrow x_0} \frac{f(x) - T_1(x)}{x - x_0} = 0.$$

Taylor polynomial: Suppose that f has n derivatives at x_0 and T_n is the polynomial of degree $\leq n$ such that

$$T_n^{(r)}(x_0) = f^{(r)}(x_0), \quad 0 \leq r \leq n.$$

How well does T_n approximate f near x_0 ?

Since T_n is a polynomial of degree $\leq n$, it can be written as

$$T_n(x) = a_0 + a_1(x - x_0) + \cdots + a_n(x - x_0)^n,$$

where $a_0, \dots, a_n$ are constants.

Differentiating yields

$$T_n^{(r)}(x_0) = r!a_r, \quad 0 \leq r \leq n.$$

We obtained a_r uniquely:

$$a_r = \frac{f^{(r)}(x_0)}{r!}, \quad 0 \leq r \leq n.$$

Therefore,

$$T_n(x) = f(x_0) + \frac{f'(x_0)}{1!}(x - x_0) + \cdots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n \\ = \sum_{r=0}^n \frac{f^{(r)}(x_0)}{r!}(x - x_0)^r.$$

Special case of Taylor series: I picked power series, I think it's right

AHMAD MSC(MATH)
Areeb2113@gmail.com



AHMAD MSC(MATH)
 Areeb2113@gmail.com

2.6. Limit Superior and Limit Inferior

39

Although each y_j occurs as a term of $\{x_n\}$, $\{y_j\}$ is not necessarily a subsequence of $\{x_n\}$, because if we write

$$y_j = x_{n_j}.$$

There is no reason to expect that $\{n_j\}$ is an increasing sequence as required in definition of subsequence.

However, it is always possible to pick a subsequence $\{n_{j_k}\}$ of $\{n_j\}$ that is increasing, and then the $\{x_{n_{j_k}}\} = \{y_{j_k}\}$ is a subsequence of both $\{y_j\}$ and $\{x_n\}$.

2.6 Limit Superior and Limit Inferior

Theorem:

If $\{s_n\}$ is bounded above and does not diverge to $+\infty$, then there is a unique real number $\bar{s}$ such that, if $\varepsilon > 0$,

$$s_n < \bar{s} + \varepsilon \quad \text{for large } n \quad (2.4)$$

and

$$s_n > \bar{s} - \varepsilon \quad \text{for infinitely many } n. \quad (2.5)$$

- If $\{s_n\}$ is bounded below and does not diverge to $-\infty$, then there is a unique real number $\underline{s}$ such that, if $\varepsilon > 0$,

$$s_n > \underline{s} - \varepsilon \quad \text{for large } n \quad (2.6)$$

and

$$s_n < \underline{s} + \varepsilon \quad \text{for infinitely many } n. \quad (2.7)$$

Proof: Proof of the first part. Since $\{s_n\}$ is bounded above, there is a number β such that $s_n < \beta$ for all n . Since $\{s_n\}$ does not diverge to $+\infty$, there is a number α such that $s_n > \alpha$ for infinitely many n . If we define

$$M_k = \sup\{s_k, s_{k+1}, \dots, s_{k+r}, \dots\},$$

then $\alpha \leq M_k \leq \beta$, so $\{M_k\}$ is bounded. Since $\{M_k\}$ is nonincreasing (why?), it converges. Let

$$\bar{s} = \lim_{k \rightarrow \infty} M_k. \quad (2.8)$$

If $\varepsilon > 0$, then $M_k < \bar{s} + \varepsilon$ for large k , and since $s_n \leq M_k$ for $n \geq k$, $\bar{s}$ satisfies (2.4).

If (2.5) were false for some positive ε , there would be an integer K such that

$$s_n \leq \bar{s} - \varepsilon \quad \text{if } n \geq K.$$

However, this implies that

$$M_k \leq \bar{s} - \varepsilon \quad \text{if } k \geq K,$$





AHMAD MSC(MATH)
Arach2113@gmail.com

5.4. Fundamental Theorem of Calculus

148

Example: If

$$f(x) = \begin{cases} x, & 0 \leq x \leq 1, \\ x+1, & 1 < x \leq 2. \end{cases}$$

Then the function

$$F(x) = \int_0^x f(t) dt = \begin{cases} \frac{x^2}{2}, & 0 < x \leq 1, \\ \frac{x^2}{2} + x - 1, & 1 < x \leq 2, \end{cases}$$

is continuous on $[0, 2]$.

We can conclude

$$F'(x) = \begin{cases} x = f(x), & 0 < x < 1, \\ x+1 = f(x), & 1 < x < 2. \end{cases}$$

$$F'_+(0) = \lim_{x \rightarrow 0+} \frac{F(x) - F(0)}{x} = \lim_{x \rightarrow 0+} \frac{(x^2/2) - 0}{x} = 0 = f(0),$$

$$F'_-(2) = \lim_{x \rightarrow 2-} \frac{F(x) - F(2)}{x - 2} = \lim_{x \rightarrow 2-} \frac{(x^2/2) + x - 1 - 3}{x - 2}$$



2.3. Monotonic Sequences

32

Example: If $s_n = [1 + (-1)^n]n$, then $\{s_n\}$ is bounded below ($s_n \geq 0$) but unbounded above.

$\{-s_n\}$ is bounded above ($-s_n \leq 0$) but unbounded below.

If $s_n = (-1)^n$, then $\{s_n\}$ is bounded.

If $s_n = (-1)^n n$, then $\{s_n\}$ is not bounded above or below.

AHMAD MSC(MATH)
Areeb2113@gmail.com

Theorem: A convergent sequence is bounded.

Proof: By taking $\varepsilon = 1$, we see that if $\lim_{n \rightarrow \infty} s_n = s$, then there is an integer N such that

$$|s_n - s| < 1 \quad \text{if } n \geq N.$$

Therefore,

$$|s_n| = |(s_n - s) + s| \leq |s_n - s| + |s| < 1 + |s| \quad \text{if } n \geq N,$$

and

$$|s_n| \leq \max\{|s_0|, |s_1|, \dots, |s_{N-1}|, 1 + |s|\}$$

for all n , so $\{s_n\}$ is bounded.

Theorem: A sequence $\{s_n\}$ converges to s if and only if every neighborhood of s contains s_n for all but finitely many n .

2.3 Monotonic Sequences

A sequence $\{s_n\}$ is *nondecreasing* if $s_n \geq s_{n-1}$ for all n , or *nonincreasing* if $s_n \leq s_{n-1}$ for all n .

A *monotonic sequence* is a sequence that is either nonincreasing or nondecreasing. If $s_n > s_{n-1}$ for all n , then $\{s_n\}$ is *increasing*, while if $s_n < s_{n-1}$ for all n , $\{s_n\}$ is *decreasing*.

Theorem:

- If $\{s_n\}$ is nondecreasing, then $\lim_{n \rightarrow \infty} s_n = \sup\{s_n\}$.
- If $\{s_n\}$ is nonincreasing, then $\lim_{n \rightarrow \infty} s_n = \inf\{s_n\}$.

Recall: The definition of supremum and infimum of a set.

Proof: Let $\beta = \sup\{s_n\}$. If $\beta < \infty$, then by definition if $\varepsilon > 0$ then

$$\beta - \varepsilon < s_N \leq \beta$$

for some integer N .

Since $s_N \leq s_n \leq \beta$ if $n \geq N$, it follows that

$$\beta - \varepsilon < s_n \leq \beta \quad \text{if } n \geq N.$$



5 marks prove from this one

Theorem: If f is integrable on $[a, b]$ and $a \leq c \leq b$, then the function F defined by

$$F(x) = \int_c^x f(t) dt$$

satisfies a Lipschitz condition on $[a, b]$, and is therefore continuous on $[a, b]$.

Proof: Consequently,

$$|F(x) - F(x')| \leq K|x - x'|, \quad a \leq x, x' \leq b.$$

Theorem: If f is integrable on $[a, b]$ and $a \leq c \leq b$, then $F(x) = \int_c^x f(t) dt$ is differentiable at any point x_0 in (a, b) where f is continuous, with $F'(x_0) = f(x_0)$.

If f is continuous from the right at a , then $F'_+(a) = f(a)$. If f is continuous from the left at b , then $F'_-(b) = f(b)$.

Proof: Since

$$\frac{1}{x - x_0} \int_{x_0}^x f(t) dt = f(x_0).$$

AHMAD MSC(MATH)
Areeb2113@gmail.com

We can write

$$\frac{F(x) - F(x_0)}{x - x_0} - f(x_0) = \frac{1}{x - x_0} \int_{x_0}^x [f(t) - f(x_0)] dt.$$

From this

$$\left| \frac{F(x) - F(x_0)}{x - x_0} - f(x_0) \right| \leq \frac{1}{|x - x_0|} \left| \int_{x_0}^x |f(t) - f(x_0)| dt \right|.$$

Since f is continuous at x_0 , there is for each $\varepsilon > 0$ a $\delta > 0$ such that

$$|f(t) - f(x_0)| < \varepsilon \quad \text{if} \quad |x - x_0| < \delta$$

and t is between x and x_0 .

Therefore, we have

$$\left| \frac{F(x) - F(x_0)}{x - x_0} - f(x_0) \right| < \varepsilon \frac{|x - x_0|}{|x - x_0|} = \varepsilon \quad \text{if} \quad 0 < |x - x_0| < \delta.$$

Hence, $F'(x_0) = f(x_0)$.

Theorem: If f is integrable on $[a, b]$ and $a \leq c \leq b$, then $F(x) = \int_c^x f(t) dt$ is differentiable at any point x_0 in (a, b) where f is continuous, with $F'(x_0) = f(x_0)$.

If f is continuous from the right at a , then $F'_+(a) = f(a)$. If f is continuous from the left at b , then $F'_-(b) = f(b)$.





Example: Prove that

$$\lim_{x \rightarrow 0^+} \sqrt{x} = 0.$$

AHMAD MSC(MATH)
break2113@gmail.com

Solution: See Lecture.

Example: Show that

$$\lim_{x \rightarrow 0^+} \left(\frac{|x|}{x} + x \right) = 1, \quad \lim_{x \rightarrow 0^-} \left(\frac{|x|}{x} + x \right) = -1,$$

Solution: See Lecture

Theorem: A function f has a limit at x_0 if and only if it has left- and right-hand limits at x_0 , and they are equal. More specifically,

$$\lim_{x \rightarrow x_0} f(x) = L$$

if and only if

$$\lim_{x \rightarrow x_0^+} f(x) = \lim_{x \rightarrow x_0^-} f(x) = L.$$





3.1 Monotone Sequences

3.1.1 Definition:

AHMAD MSC (MATH)
Ahmad21113@gmail.com

- (i) A sequence $(a_n)_{n \geq 1}$ said to be **monotonically increasing**, if $(a_n)_{n \geq 1} \geq a_n \forall n \geq 1$.
- (ii) A sequence $(a_n)_{n \geq 1}$ said to be **monotonically decreasing**, if $(a_n)_{n \geq 1} \leq a_n \forall n \geq 1$.

We can describe now the completeness property of the real numbers.

3.1.2 Completeness property

Every monotonically increasing sequence which is bounded above is convergent.

3.1.3 Theorem:

If $(a_n)_{n \geq 1}$ is monotonically decreasing and is bounded below, it is convergent.

Proof:

Follows from the following facts

- (i) $(a_n)_{n \geq 1}$ monotonically decreasing if and only if $(a_n)_{n \geq 1}$ monotonically increasing.
- (ii) $(a_n)_{n \geq 1}$ bounded below, if and only if $(a_n)_{n \geq 1}$ is bounded above.
- (iii) $(a_n)_{n \geq 1}$ is convergent if and only if $(a_n)_{n \geq 1}$ is convergent.

3.1.4 Examples:

- (i) Sequence $(2n)_{n \geq 1}$ is monotonically increasing and is not bounded above.
- (ii) Sequence $(\frac{1}{n})_{n \geq 1}$ is monotonically decreasing and is not bounded below, say by 0.
- (iii) Sequence $((-1)^n)_{n \geq 1}$ is neither monotonically increasing nor decreasing.

- (iv) Let $a_1 = 1$, $a_n = \frac{3a_{n-1} + 2}{6} \forall n \geq 2$. As shown in problem 2, module 0, $a_n \geq \frac{2}{3} \forall n$ and

